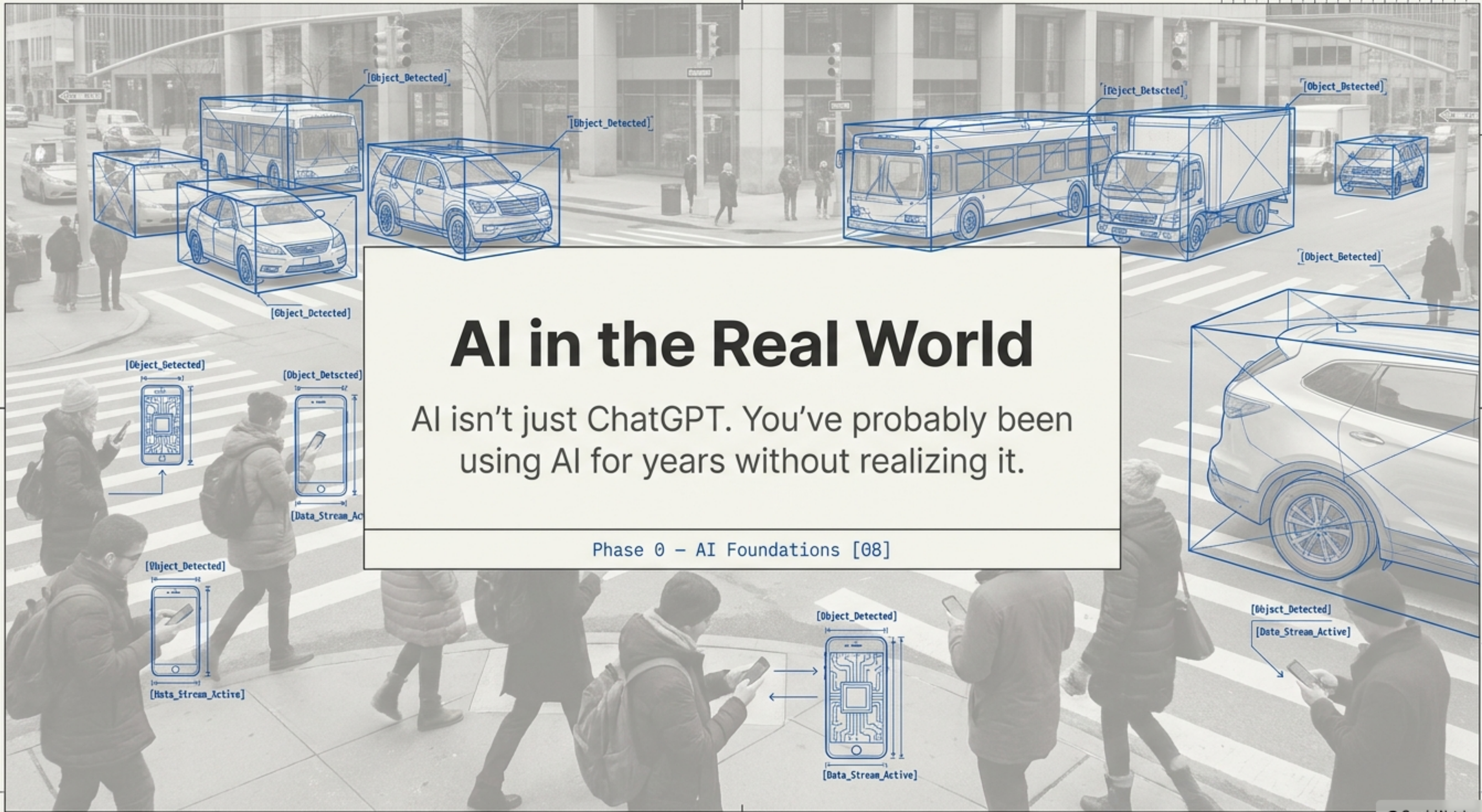
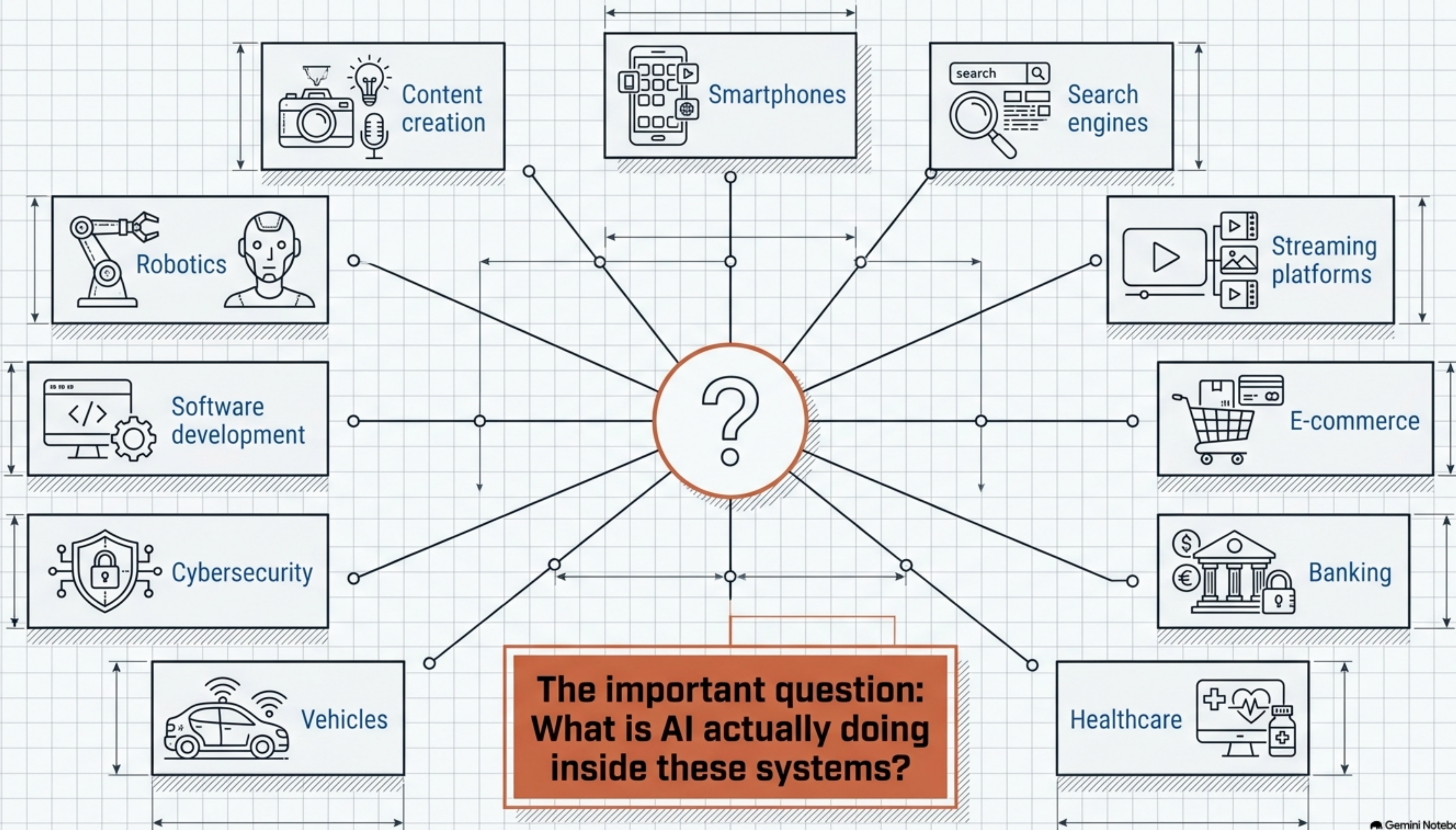


AI in the Real World

AI isn't just ChatGPT. You've probably been using AI for years without realizing it.

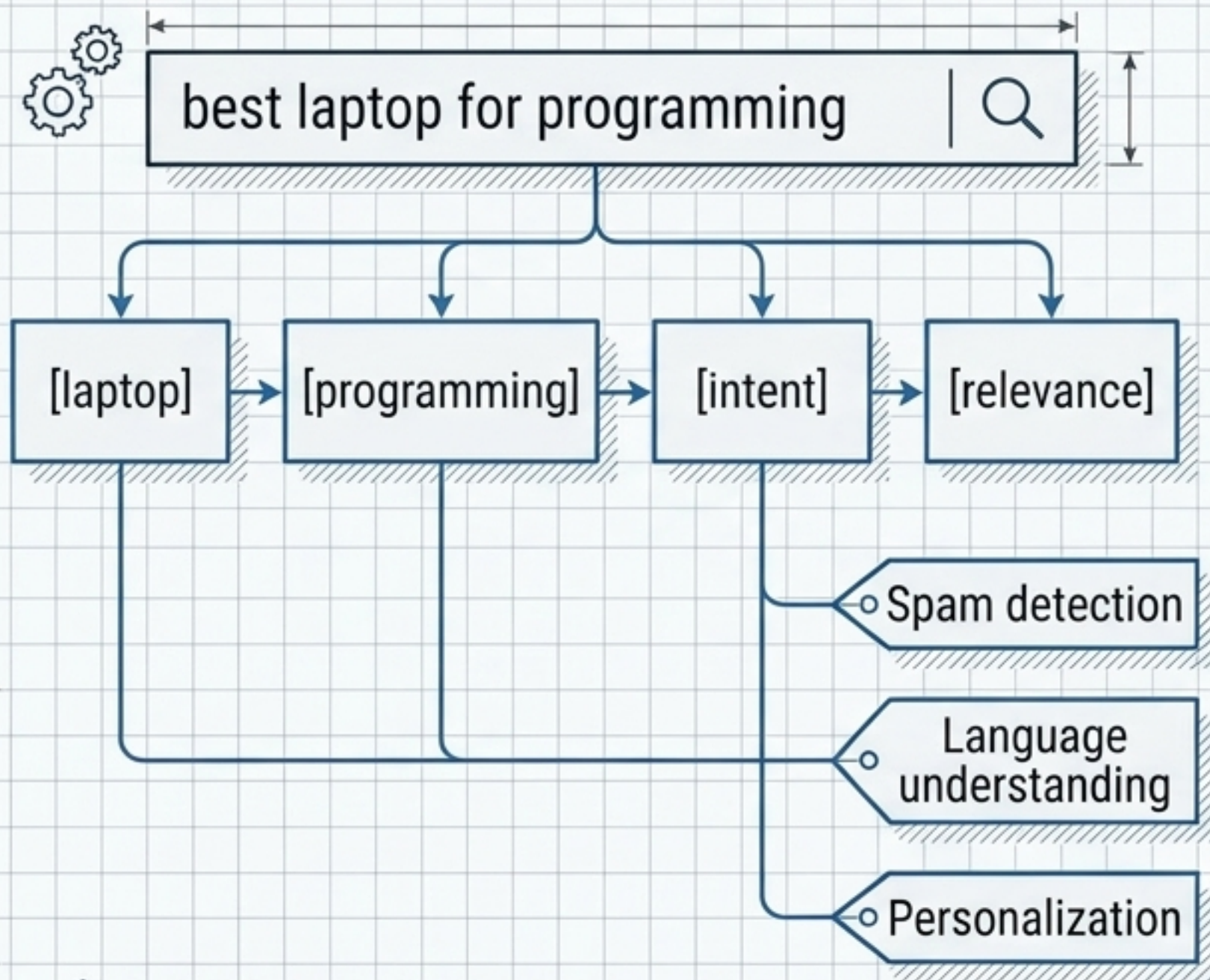
Phase 0 – AI Foundations [08]





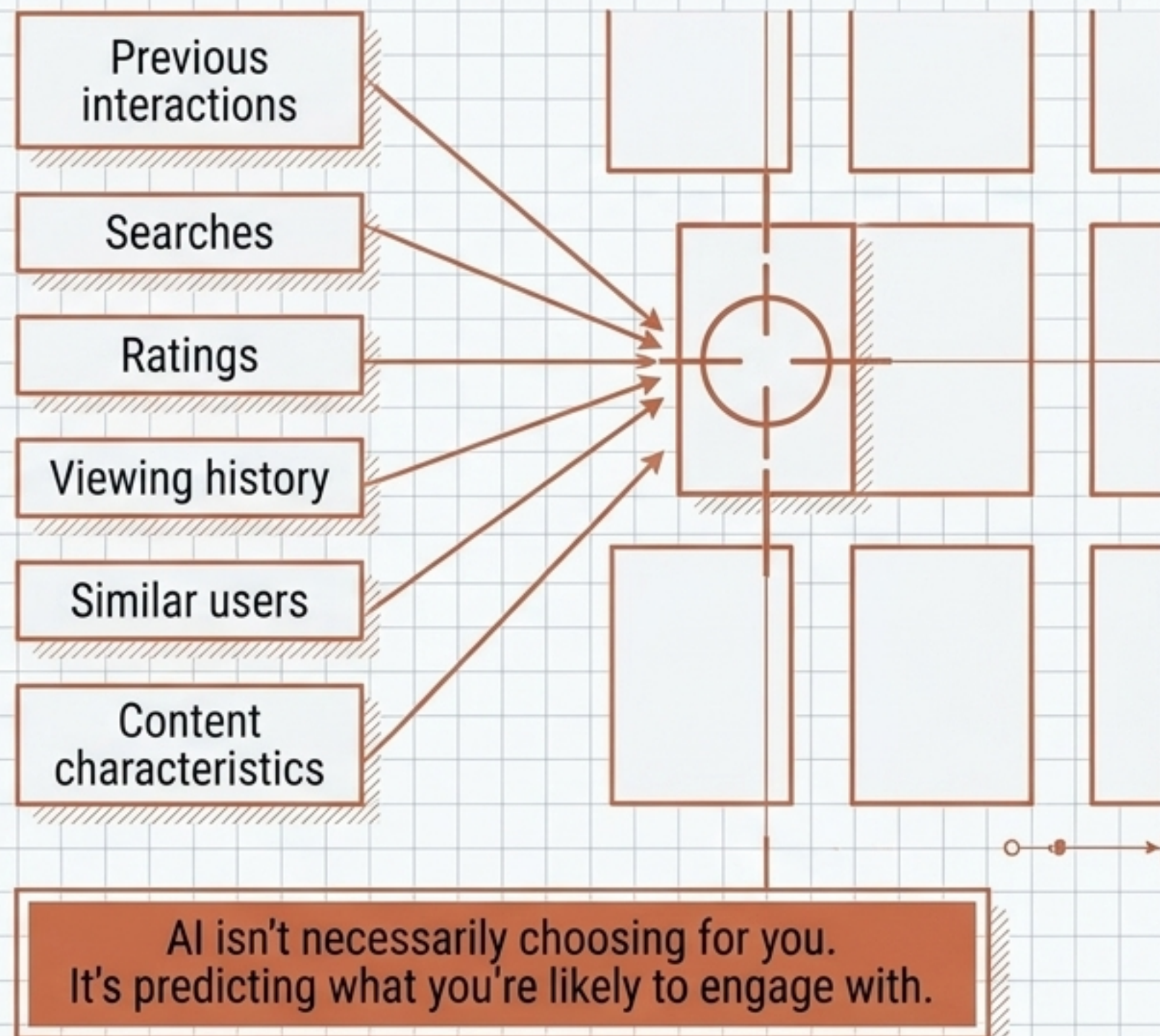
Search Engines

Understanding & Ranking



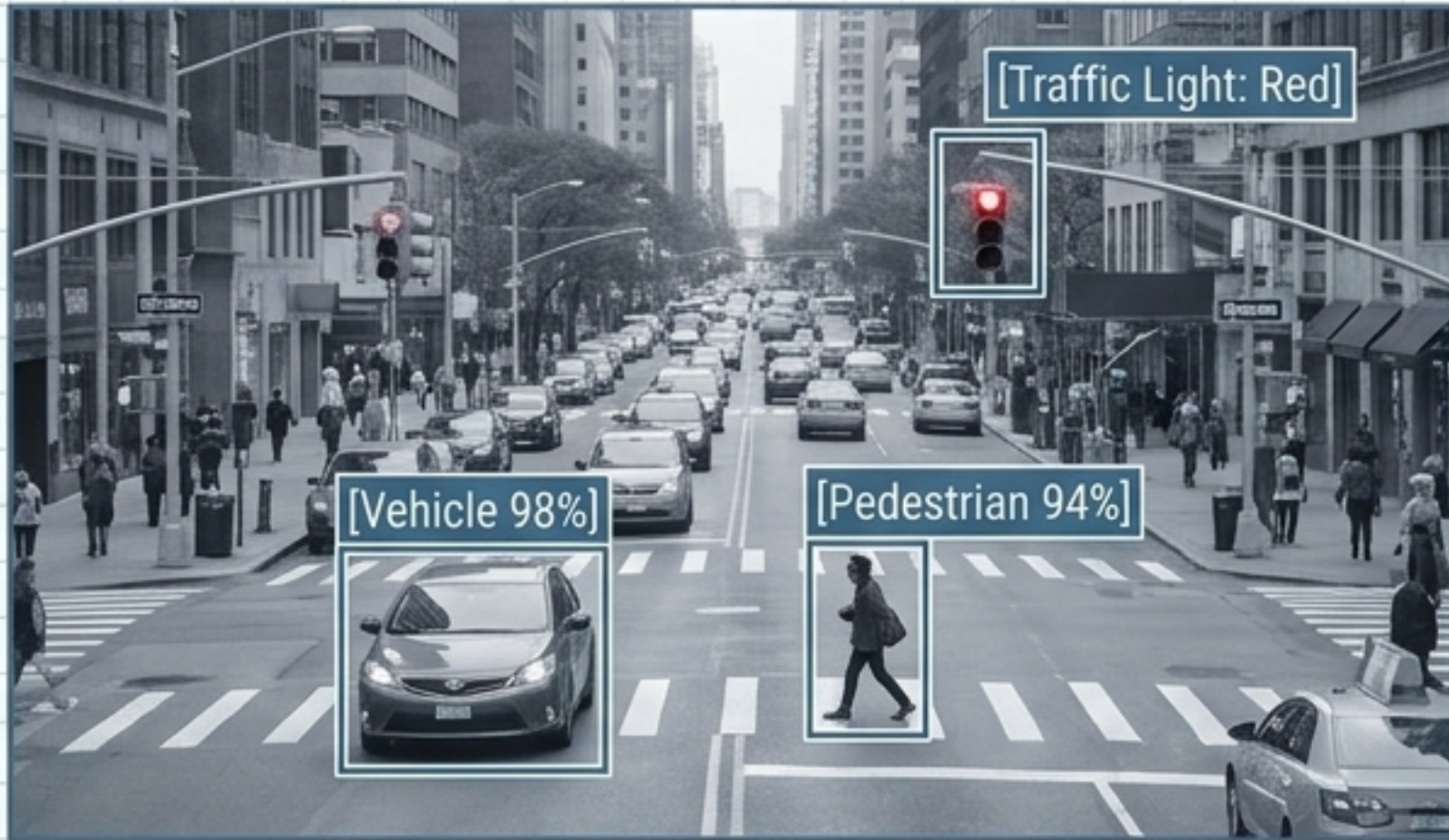
Recommendation Systems

Predicting the next click



Computer Vision

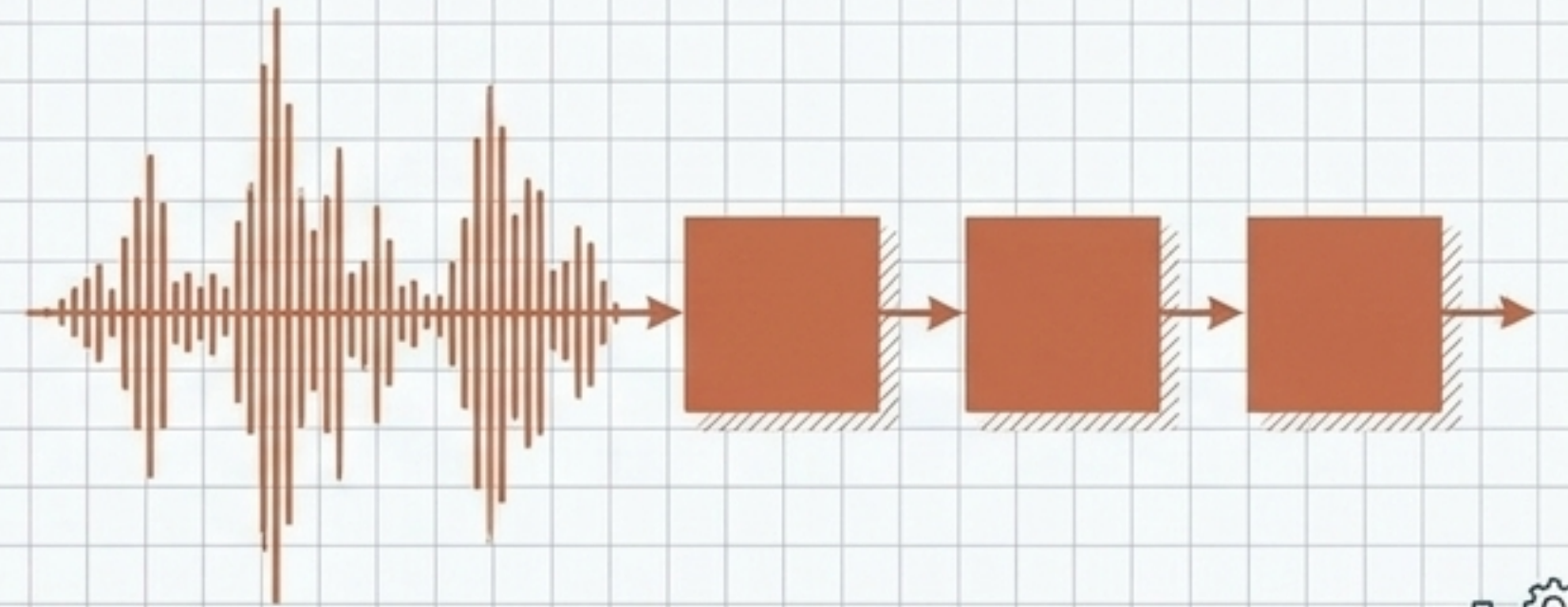
Interpreting visual information



Face recognition	OCR
Object detection	Autonomous vehicles
Medical imaging	Security systems
Quality inspection	

Speech & Voice AI

Processing human speech

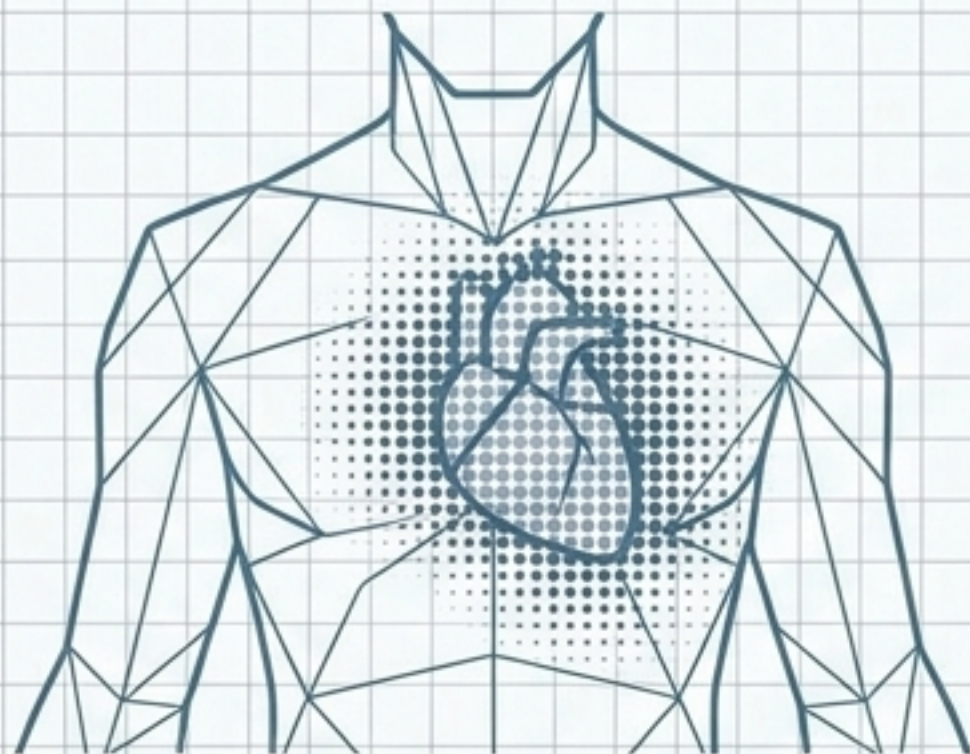


Voice assistants	Voice search
Speech-to-text	Translation
Call transcription	Voice generation

Much more than simply converting audio to text.

Healthcare

Assistance & Risk Prediction



Medical image analysis

Drug discovery

Clinical decision support

Patient monitoring

Documentation

AI is a tool for assistance, not an automatic replacement for professional medical judgment.

Finance

Anomaly Detection



This transaction doesn't look like the customer's usual behavior.

Fraud detection

Credit risk analysis

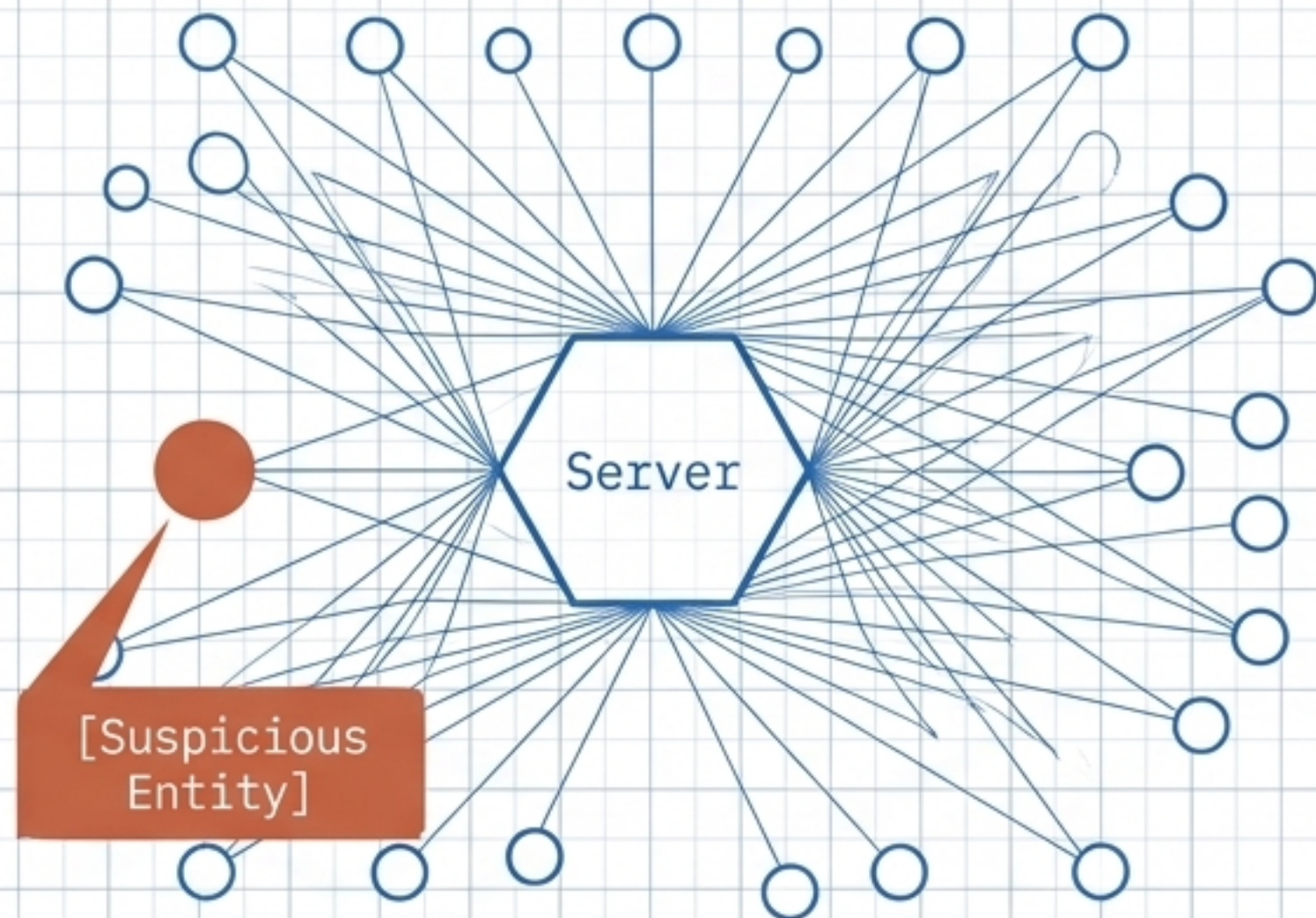
Algorithmic trading

Forecasting

Anti-money laundering

Cybersecurity

Pattern recognition at scale



Malware/Phishing detection

Network anomaly detection

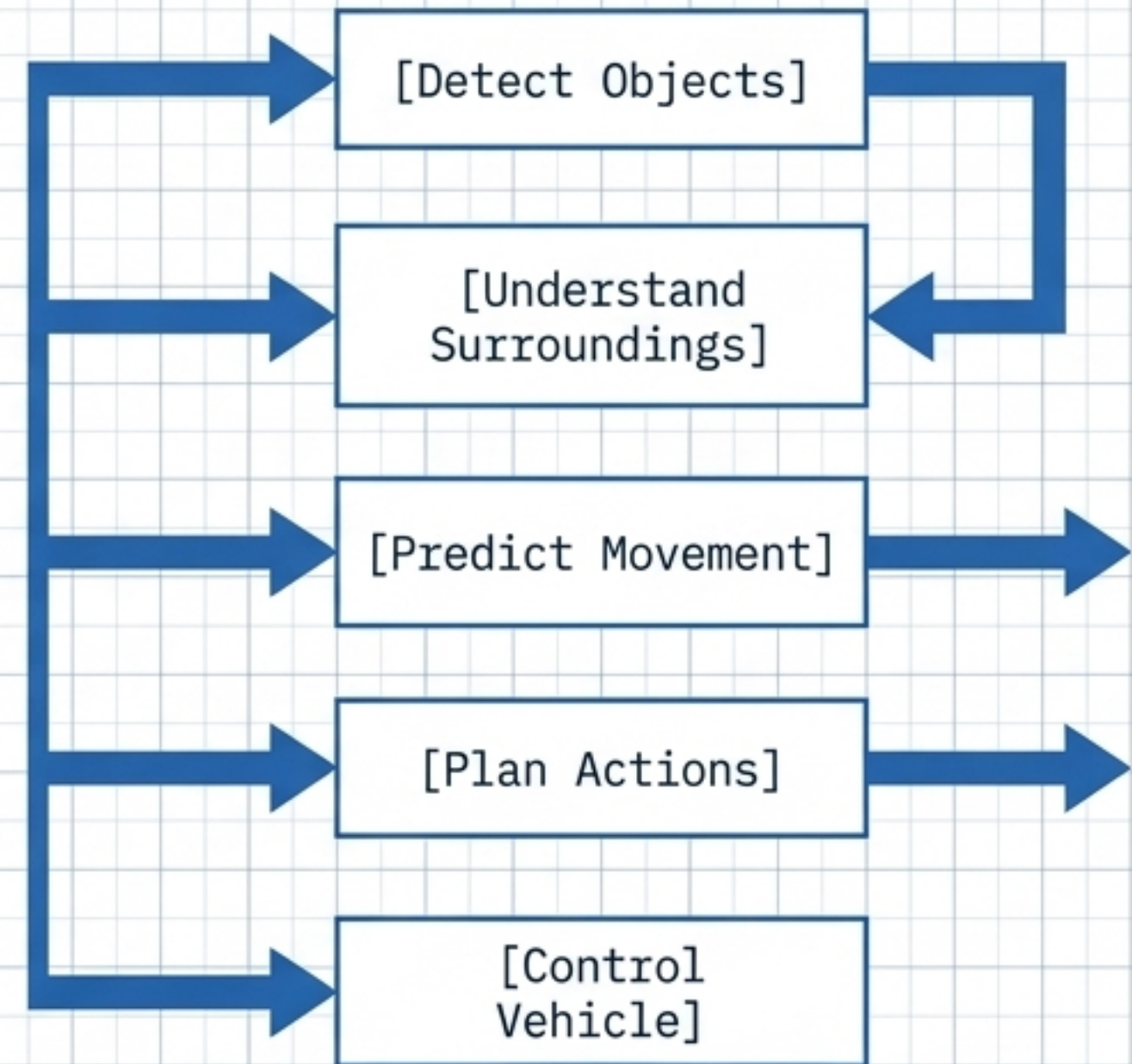
User behavior analysis

Threat classification

Detects patterns that are difficult to manually monitor at scale.

Autonomous Systems

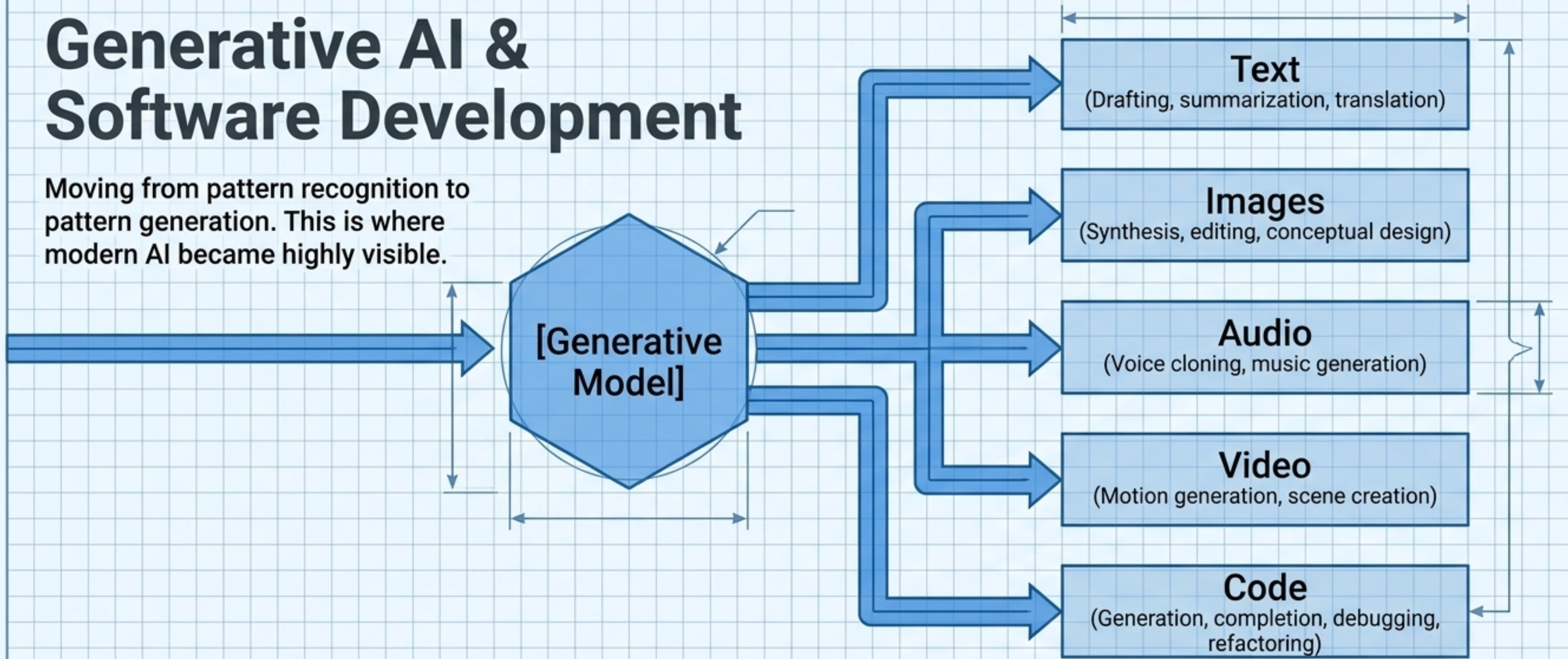
Closed-loop processing



This is a composite system, not just one AI model.

Generative AI & Software Development

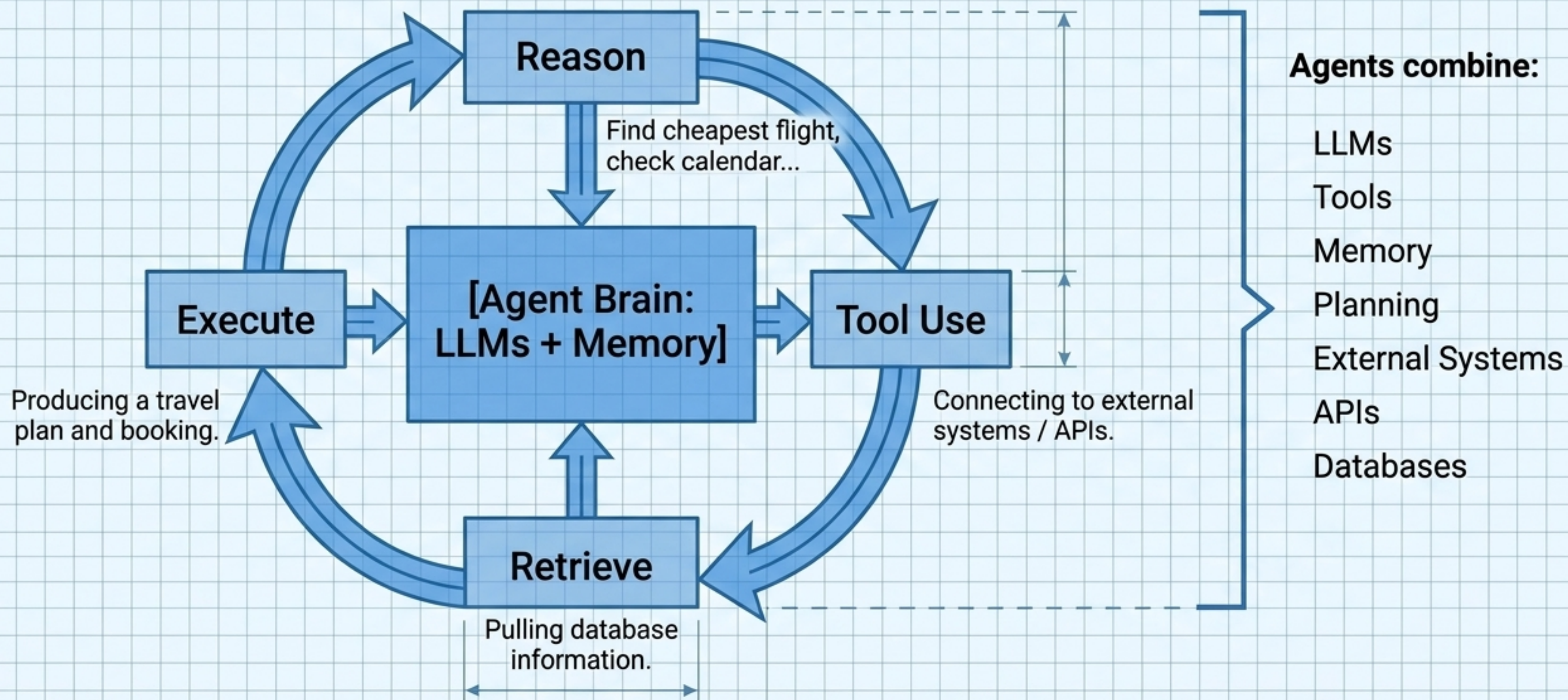
Moving from pattern recognition to pattern generation. This is where modern AI became highly visible.



AI can generate the code. The developer still needs to understand and verify it.

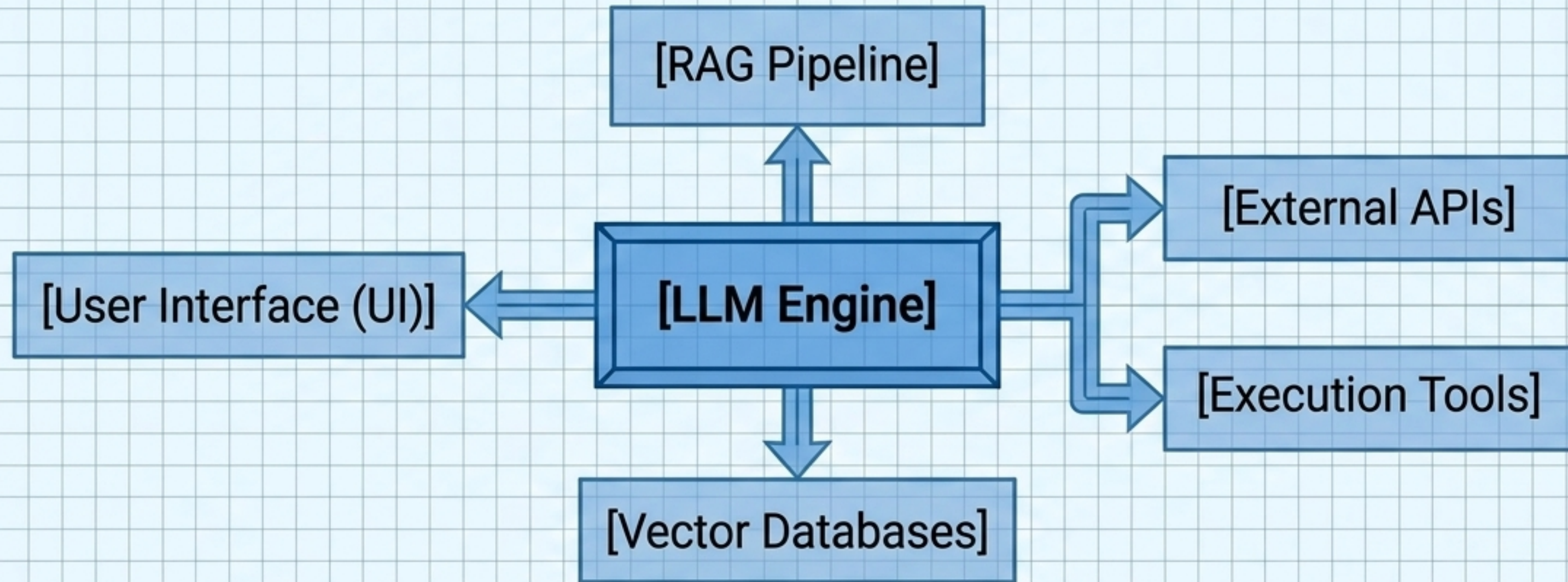
The Next Frontier: AI Agents

Moving beyond basic Input -> Answer



The Anatomy of a System

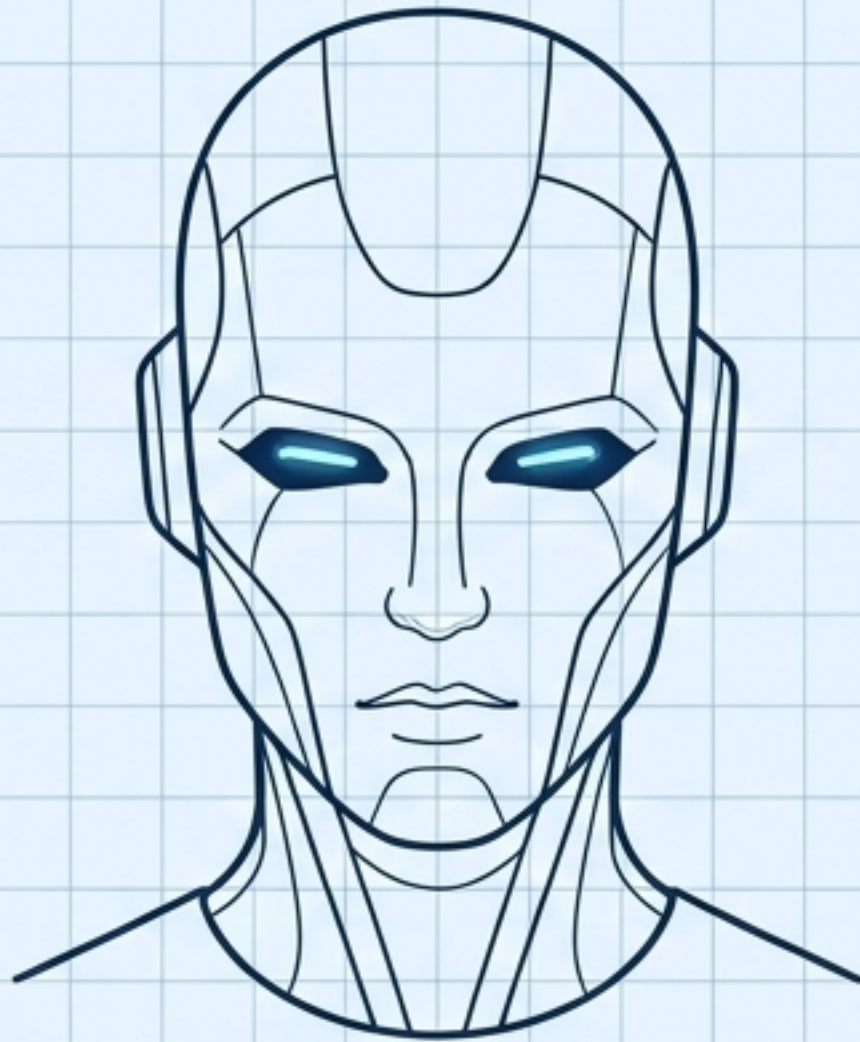
One AI system requires many technologies working together.



Real-world AI is rarely just one model. Deploying AI is about building a reliable system around it.

Expectation vs. Reality

Expectation



User: "AI is going to replace everyone."

Reality



Meanwhile, the AI system trying to call an API:

The Reality Check

AI is excellent at...

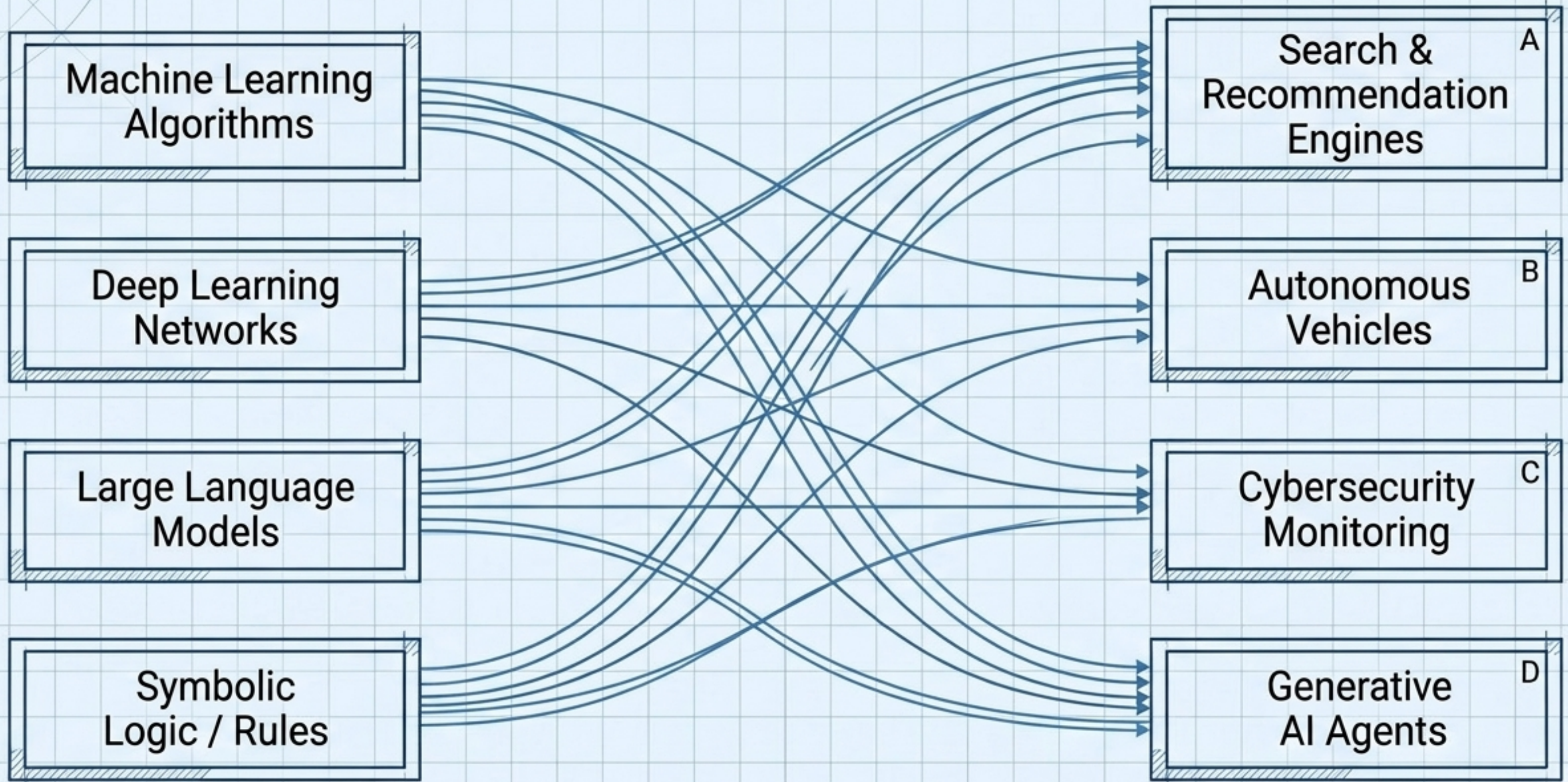
+	Scale
+	Speed
+	Pattern recognition
+	Automation
+	Repetitive tasks
+	Large-scale computation
+	Content generation

AI can struggle with...

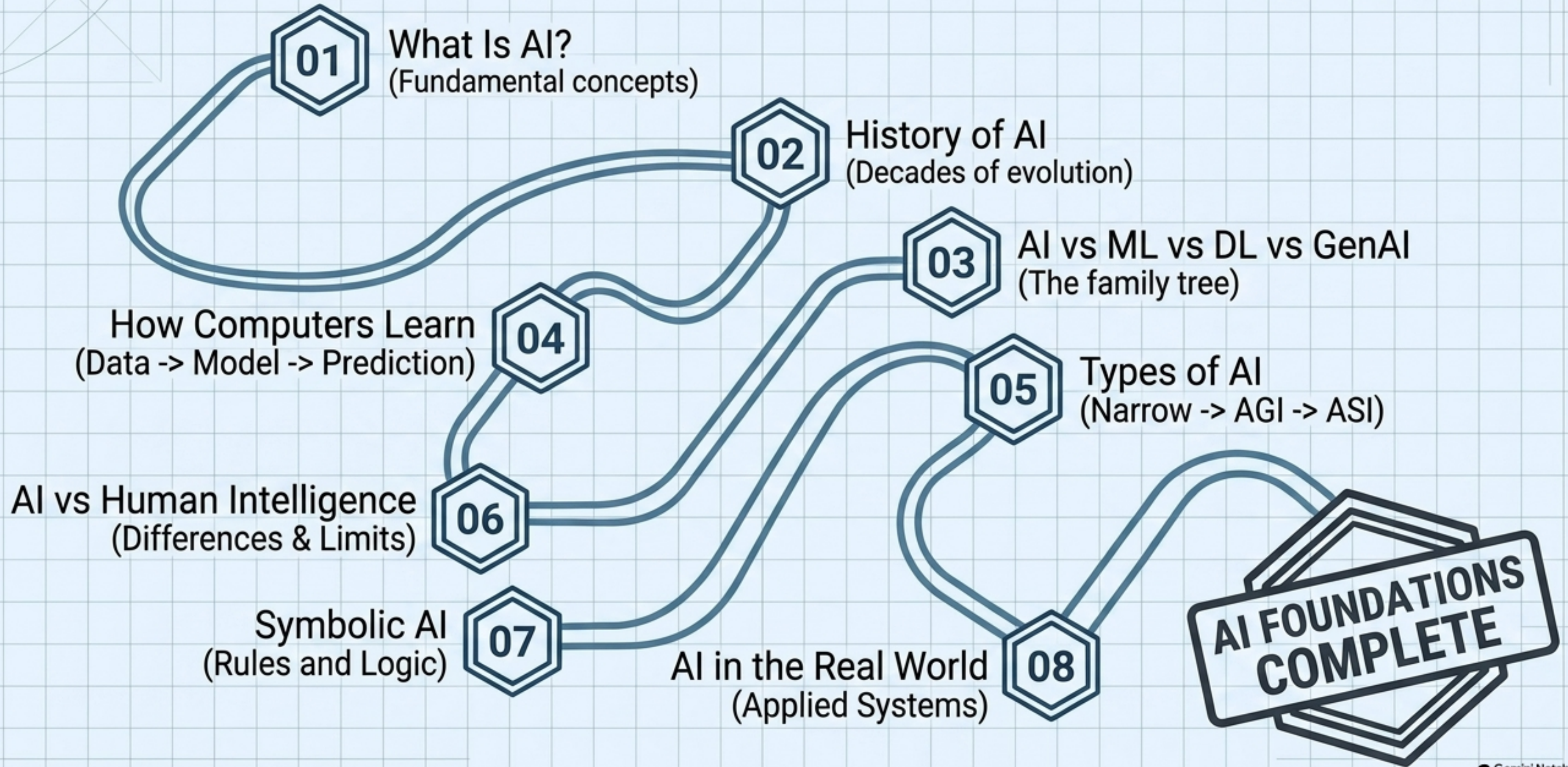
!	Hallucinations
!	Bias
!	Poorly specified objectives
!	Unexpected situations
!	Reliability
!	Explainability
!	Security and privacy risks

Deploying AI isn't just about making a model that works.
It's about building a reliable system around it.

The Complete Picture



Phase 0 Complete: The Journey



Final Phase 0 Takeaway

AI isn't magic. It isn't one single technology.
It is an enormous field built from:

**[Mathematics] + [Algorithms] +
[Data] + [Computing] + [Research]
= Artificial Intelligence**

We have evolved from theoretical concepts into
the integrated systems that run our modern world.

RETAIN CURIOSITY.

The foundations are set. You now know what AI is, how it learns, and where it lives.

INITIATING NEXT PHASE...